CS2060 Programming with C - Ch. 5 Assignment

1 Instructions

1. Submit this assignment on Canvas.
2. Solve each question in a separate (.c) file.
3. Print screen of the execution of each question (as evidence that your code is working).
4. It is highly recommended to run your code (for each question) on the test cases and print screen the results each time.
5. For each question, collect the .c file and its print screens into a folder.
6. Create a folder for the whole assignment.
7. Place all question folders into the assignment folder.
8. Compress the assignment folder into one zipped folder.
9. Upload this zipped folder to Canvas.
10. It is your own responsibility to check that you uploaded the assignment correctly (you can open the submission web page on Canvas and download your submission to check everything is OK).
11. If the Canvas is down and the deadline is too close, you can send the assignment via email to me and the TA with a print screen of the Canvas page that is down (as evidence that the Canvas website is down).

2 Questions

1. Write a C program to perform the following:
    - Asking the user to enter using the keyboard an integer positive number.

- You should implement a function called reverseDigits that takes that number and returns the number with its digits reversed. For example, given the number 7631, the function should return 1367.

```
Please, enter a positive integer number: -1
Invalid Input
```

```
Please, enter a positive integer number: 0
Invalid Input
```

```
Please, enter a positive integer number: 1
The number after reverse its digits is: 1
```

```
Please, enter a positive integer number: 90
The number after reverse its digits is: 9
```

```
Please, enter a positive integer number: 123
The number after reverse its digits is: 321
```

```
Please, enter a positive integer number: 7631
The number after reverse its digits is: 1367
```

2. Write a C program to perform the following:
   - Asking the user to enter using the keyboard 2 integer numbers.
   - You should define 2 functions (gcdNonRecursive and gcdRecursive):
     - The gcdNonRecursive function takes the 2 numbers and calculates the Greatest Common Divisor (GCD) of the 2 numbers non-recursively.
     - The gcdRecursive function takes the 2 numbers and calculates the Greatest Common Divisor (GCD) of the 2 numbers recursively.

```
Please, enter two integer numbers: 0 50
The greatest common divisor is (using a non-recursive function): 50
The greatest common divisor is (using a recursive function): 50
```

```
Please, enter two integer numbers: 0 -50
The greatest common divisor is (using a non-recursive function): 50
The greatest common divisor is (using a recursive function): 50
```

```
Please, enter two integer numbers: 42 120
The greatest common divisor is (using a non-recursive function): 6
The greatest common divisor is (using a recursive function): 6
```

```
Please, enter two integer numbers: 150 35
The greatest common divisor is (using a non-recursive function): 5
The greatest common divisor is (using a recursive function): 5
```